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Experimental investigation on biofouling marine bryozoans in reefs

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Abstract

The natural colonisation of organisms on submerged sea surfaces is known as marine biofouling, and it presents challenges to both the environment and management organisations. Bryozoans play an important role in the creation of ecosystems and the preservation of biodiversity. *Parasmittina projecta* and *Celleporaria aperta* were found to be the most common species of marine bryozoans that encrusted on ceramic and terracotta plates, according to the present study conducted in the Gulf of Mannar Marine Biosphere Reserve at Pamban, Vedalai, Manouli, and Ervadi Islands. On the other hand, *Parasmittina tubula* and *Electra pilosa* were found to be less common. No bryozoan colonies developed on the metal panels; however, these panels exhibited substantial abiotic corrosion caused by prolonged exposure to seawater, which may have inhibited bryozoan settlement. The present investigation lasted for a year and identified ten different bryozoan species with *P. projecta* being the most prevalent. Submerged panels formed microhabitats that supported a variety of marine organisms, while competent species overgrew and displaced less competent ones. This study highlights the positive role of biofouling bryozoans in sustaining biodiversity, facilitating macro-fouler settlement, acting as reef-forming agents, and providing benefits for fisheries, underwater recreational activities, ecotourism, and disaster management.

Keywords: Bryozoa, Gulf of Mannar, Macro-fouler settlement, Marine biofouling, *Parasmittina projecta*

1 Introduction

Marine biofouling is a natural phenomenon involving the colonization of organisms on submerged surfaces in the marine environment, which causes major management issues and has environmental consequences [19]. Biofouling poses considerable threat to marine vessels as it significantly increases hydrodynamic friction, which leads to increased propulsion drag and associated fuel consumption [25]. It can also impede intake pipes of cooling systems causing severe obstruction in water passages [25]. More importantly, biofouling allows introduction of marine species known as Introduced Marine Pests (IMPs), which represent one of the most serious threats to the world's oceans [2]. Biofouling commences with the formation of a biofilm, 1 mm in thickness, and gradually develops through successional processes [3]. Filamentous green, red and brown algae adhere to the surfaces coated with biofilm (sticky substances) secreted by



bacteria or unicellular algae, thereby facilitating the settlement of sessile animals such as bryozoans, serpulid tube worms and barnacles [19]. Together, these organisms act as pioneering macro-fouling species creating micro-crevices and suitable substrates that attract mobile benthic and epi-benthic animals, such as crabs, brittle stars, and small fish as well as parasites, commensals and pathogens, transforming the submerged surfaces into a functional ecosystem [3].

Bryozoans are an important fouling community as they form extensive crusts, which provide a safe substratum for the attachment of other macro-fouling species. Bryozoans begin their life as free-swimming larvae developed from fertilized eggs [20]. The larval stage is critical in fouling, as it secretes a highly adhesive polysaccharide substance that enables the larva to attach to surfaces [20]. This polysaccharide substance is rapidly mineralized to form a thick solid crust [50]. Fouling bryozoan species are tolerant to a wide range of environmental conditions, enabling them to establish and integrate into the fauna in new areas [19]. There are numerous organisms capable of encrusting and surviving on wood, metal, ceramic, or terracotta surfaces found on the hulls of boats and ships, as well as on docks and harbours that come in contact with seawater, and bryozoans are one among them [63]. Bryozoan larvae also tend to exhibit habitat preferences based on exposure to light, water movement, substrate texture, colour and the physical or chemical properties of the surface [33, 44], which affect its fouling behaviour.

To date, no experimental investigation has been conducted at the Gulf of Mannar and Palk Bay to study the role of marine bryozoans as biofoulers. The present study gives an overview of the role of marine bryozoans as reef-forming agents, contributing to community structure and supporting biodiversity. Habitat preferences, seasonality of settlement, and abundance of bryozoan species were assessed through both short-term and long-term studies conducted at four different locations in the Gulf of Mannar Marine Biosphere Reserve, namely Pamban Island, Vedalai, Manouli Island and Ervadi, between June 2020 and June 2021. Although bryozoans in the study area were previously not regarded as ecologically significant, the present study highlights their substantial ecological value.

2 Materials and methods

2.1 Study area

2.1.1 Site selection criteria

The four study sites were selected based on a set of ecological, oceanographic and logistical criteria relevant for assessing bryozoan recruitment and biofouling patterns in the Gulf of Mannar. The primary criteria included:

Documented Marine Biodiversity: All selected sites are known to harbour rich marine flora and fauna, including coral assemblages, macrofoulers, and diverse invertebrate communities. This ensured the presence of natural larval pools necessary for colonisation on artificial panels.

Minimal Anthropogenic Disturbance at 5 m Depth: Although coastal activity is present in the region, the selected panel deployment depth (5 m) ensured reduced interference from fishing gear, boat traffic and tourism providing relatively undisturbed conditions for long-term colonisation.

Accessibility for Periodic Monitoring: Sites were selected within safe diving range to allow repeated retrieval, replacement and examination of plates during both short-term (6-month) and long-term (1-year) studies.

Protected Status Within the Biosphere Reserve: Since the aim of the study was to understand natural fouling processes, choosing locations within the Gulf of Mannar Marine Biosphere Reserve ensured that panels were placed in ecologically regulated zones with consistent conservation frameworks.

2.1.2 Pamban Island (site 1)

Pamban Island (9.276881° , 79.202691°), commonly known as Rameswaram Island due to the pilgrimage centre of Rameswaram, is situated at the centre of the islands and east of Vedalai (Fig. 1). It is the largest island in Tamil Nadu with an area of 96 km^2 , located in between India and Sri Lanka. Pamban Island consists of a few hillocks and other elevated physical features but apart from this most of the Island is covered with white sand. The island has a rich marine biodiversity as it lies between the Palk Bay to the northeast and the Gulf of Mannar to the south [39].

2.1.3 Vedalai (site 2)

Vedalai (9.254264° , 79.106386°) is a fishing village with 13 km^2 area in Ramanathapuram district, Tamil Nadu (Fig. 1). It is situated between the Palk Bay to the north and Gulf of Mannar to the south. The shoreline of Vedalai is rich in marine biodiversity [55].

2.1.4 Manouli Island (site 3)

Manouli Island (9.21564° , 79.12834°) is among the seven Mandapam group of islands in Gulf of Mannar Marine Biosphere Reserve (Fig. 1). It has an area of 0.259 km^2 situated

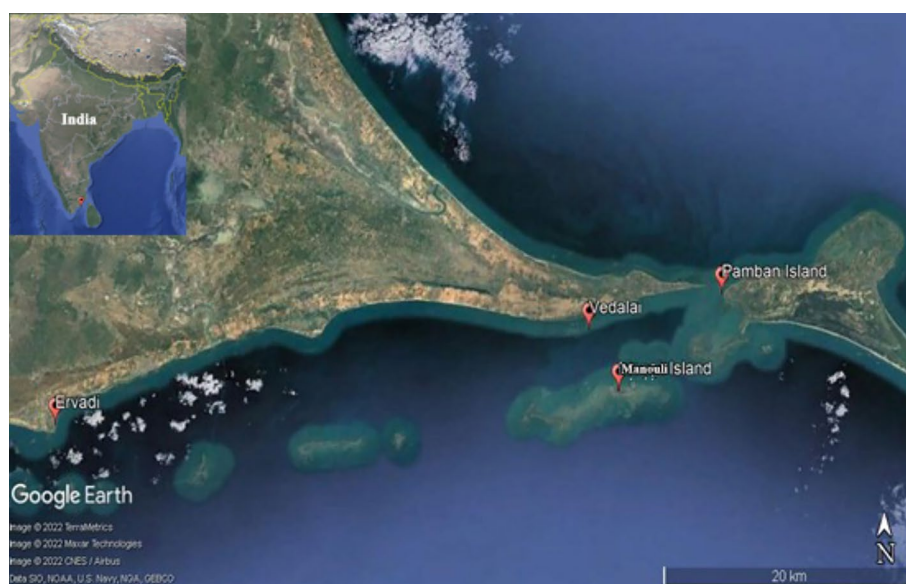


Fig. 1 Map showing the study area for experimental investigation

on the narrow peninsula projecting from the southeast coast of India. Manouli Island is located between Hare Island to the north and Manouli-Putti Island to the south. The inshore region of Manouli Island is rich in flora and fauna, which is well protected as it lies within the Biosphere Reserve [55].

2.1.5 *Ervadi (site 4)*

Ervadi (9.194329°, 78.719649°) is a village located on the coast between Keelakarai and Valinokkam in Ramanathapuram District, Tamil Nadu (Fig. 1). Ervadi, being a coastal village along Gulf of Mannar it is rich in marine biodiversity [55].

2.2 Methodology

The experimental investigation to identify the habitat preferences of marine bryozoans were conducted using static immersion panels secured to the sea floor at 5 m depth. The static immersion panel for bryozoan recruitment was constructed as a roof shape, using two 3ft by 3ft metal squares joined on top at 45° angulation and separated by a distance of 2ft at the base. The panel was secured to the sea floor at 5 m depth using metal pegs and metal wires. Plates (12inch×12inch) made of wood, metal, ceramic (denser, less porous, smoother and more impermeable surface compared to terracotta) and terracotta were fixed on to the panel using a single metal peg in the centre as seen in the figure (Fig. 2). The plates were equidistant from each other to avoid interference from the adjacent plates. In total, eight static immersion panels were employed, two per location: one for the short-term study and one for the long-term study. The study was designed as a comparative, observational colonisation experiment. At each site, one short-term and one long-term immersion panel were deployed to assess temporal patterns of bryozoan settlement. These deployments were not intended as statistical replicates but as parallel observations under identical environmental conditions.

Short-term panels were exposed for 6-months, and this was repeated twice from June 2020 to January 2021 and from January 2021 to June 2021. For the second short-term study, the initial set of plates were removed and replaced without dismantling the static immersion panel. Long-term panels remained submerged for full year from June 2020 to June 2021, at the same location without any anthropogenic disturbances. The experiment was conducted simultaneously at the sites.

The plates were analysed for biofoulers using the quadrat method by dividing the plate into 4×4 grid (Fig. 3). Algae, sediment and green layer (early-stage algal deposition, mainly consisting of microalgae and fine green algal filaments) were estimated by area while the bivalves, barnacles, tubeworms, and bryozoan colonies (average sized colony of 100 zooids) were measured by point count. Biofouling was expressed as presence/absence and the number of colonies of bryozoans.

Further, to complement colony counts and to provide a quantitative measure of substrate occupation, total bryozoan percentage cover was estimated for each panel using photographic point-count analysis. High-resolution photographs (6 nos. for each plate) of all retrieved plates were taken immediately after recovery, ensuring consistent orientation and scale. A grid of evenly spaced points was overlaid on each plate image using image analysis software (ImageJ Version 1.54) [1]. At each point, the presence or absence of bryozoans was recorded, irrespective of species identity. Percentage cover was



Fig. 2 Static immersion panels for bryozoan recruitment. C: ceramic, T: terracotta, W: wood, M: metal

calculated as the proportion of points intersecting bryozoan colonies relative to the total number of grid points, expressed as a percentage. This method estimates the total bryozoan cover per panel, rather than the area of individual colonies, and provides a robust measure of overall substrate occupation and ecological dominance.

Statistical analyses were conducted to examine patterns of bryozoan settlement and substrate occupation using total percentage cover data. Given the exploratory nature of the study and the absence of true replication, analyses were descriptive and comparative rather than strictly inferential. Bryozoan percentage cover was summarised as mean $\pm$ standard deviation for each substrate type, study period, site, and panel orientation (front vs reverse). Data normality was assessed using the Shapiro–Wilk test; as assumptions of normality were not met, non-parametric tests were applied. Differences among substrate types were evaluated using the Kruskal–Wallis test, while differences between panel orientations were assessed using the Mann–Whitney U test. Seasonal

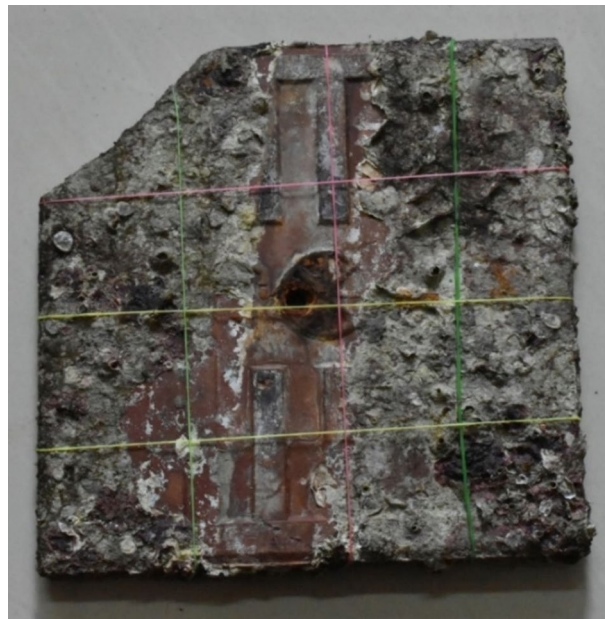


Fig. 3 Quadrat method analysis of the plate

differences between short-term study periods were examined using non-parametric comparative test (Kruskal–Wallis H test). Exploratory multivariate patterns in bryozoan settlement across substrates, seasons, and sites were visualised using principal component analysis (PCA) based on total percentage cover. Ordination results were used for pattern visualisation only. All analyses were performed using IBM SPSS Statistics for Windows (Version 23.0) [18], and results are interpreted cautiously to support ecological interpretation.

Bryozoan colonies from each plate were carefully removed using fine forceps and preserved in 70% ethanol for laboratory examination. Identification was carried out using a stereomicroscope ($10\times$ – $40\times$ magnification) to study colony morphology, zooid structure, orifice shape, operculum characteristics, avicularia, ovicells, and surface ornamentation. For species requiring finer resolution, selected colonies were examined under a compound microscope after careful cleaning with a soft brush to remove debris. Taxonomic identification was performed following standard bryozoan identification guides and regional monographs described in the literature review section of Venkatraman et al. [56] and Tilbrook and Grave [54]. Identifications were verified against descriptions in the WoRMS [62] database to ensure taxonomic accuracy and updated nomenclature. Also, selected bryozoan colonies were initially examined under a stereomicroscope (Leica EZ4). Selected, identified colonies were then coated with a gold–palladium layer under vacuum conditions, and scanning electron micrographs were obtained using a ZEISS EVO 18 Special Edition scanning electron microscope. Image acquisition and processing were performed using Smart SEM version 5.09.

This study was designed as a passive, observational colonisation experiment rather than a treatment–control design. No artificial treatments were imposed on the substrates, and all panels were exposed to identical natural environmental conditions at each site. Therefore, the ceramic, terracotta, wood, and metal plates served as comparative

substrates within the same environmental context, functioning as internal controls for each other. Including a 'no-substrate' control is not applicable in marine colonisation experiments, as natural settlement can only be assessed on a physical substrate [31].

Rainfall has a direct effect on the temperature and salinity of water, which in turn influences the coastal aquatic ecosystems. Therefore, the short-term study period was classified mainly based on the rainfall pattern in the region, as study 1 period receiving the most rainfall and study 2 period, receiving the least rain fall. Short-term study 1 was conducted during June to December which includes July, August and September South West monsoon, also known as pre-monsoon and in October, November and December North East monsoon also called Monsoon since it brings heavy rainfall. On the contrary, the short-term study 2 was conducted during January to May, which is (January, February and March) the Post-Monsoon and (April, May and June) the summer season [42].

3 Results

3.1 Short-term study

Panels used in the short-term study were submerged in the sea for 6-months period from June 2020 to January 2021 and January 2021 to June 2021, at four different locations, Site 1—Pamban Island (GMF 1), Site 2—Vedalai (GMF 2), Site 3—Manouli Island (GMF 3) and Site 4—Ervadi (GMF 4). After the period of 6 months, the plates were retrieved and analysed for biofoulers and the following observations were noticed.

3.1.1 Short-term study 1 (June 2020–January 2021)

Seven encrusting bryozoans species were identified only on the ceramic plate and terracotta plates. *Parasmittina projecta* [36] and *Celleporaria aperta* [16] were the most abundant irrespective of the location. *Parasmittina tubula* [22] *Parasmittina egyptiaca* [59], *Hippopodina feegeensis* [8], and *Pleurocodonellina signata* [58] were observed very rarely as a small colony or a patch on the submerged panels (Fig. 4). *Electra pilosa* [26] from Gulf of Mannar was found to dominate the rough surface of the terracotta plate at site 1 in the short-term study (Fig. 4).

3.1.1.1 Pamban Island (site 1) The front side of the Ceramic plate which has a rough texture, was encrusted with 32 colonies of *P. projecta* and 1 colony of *H. feegeensis*. A higher number of colonies were observed on the upper half of the plate than the lower half, while the reverse side of the plate which was smooth textureed, was completely devoid of any bryozoan colony (Fig. 5 PCE, PCR). The front side of the Terracotta plate which is smooth in texture, was sparsely encrusted by bryozoans forming small colonies of 2 cm to 5 cm in diameter. It composed of four colonies of *E. Pilosa*, six colonies of *P. projecta*, three colonies of *P. tubula* and six colonies of *P. egyptiaca*. The reverse side which was rough in texture was extensively coated with large colonies of *E. pilosa*. A large number of colonies were noticed on the upper half of the plate (Fig. 5 PTF, PTR). No bryozoan colonies were seen on both surfaces of the wooden block. The front side was more or less barren and the reverse side was completely coated with sand and salt deposits (Fig. 5 PWF, PWR). Both surfaces of the Metal Sheet were completely coated with algae and salt deposits but no bryozoan colony was seen. Since the metal sheet was not painted, it had started rusting (Figs. 5 PMF, PMR; 6i).

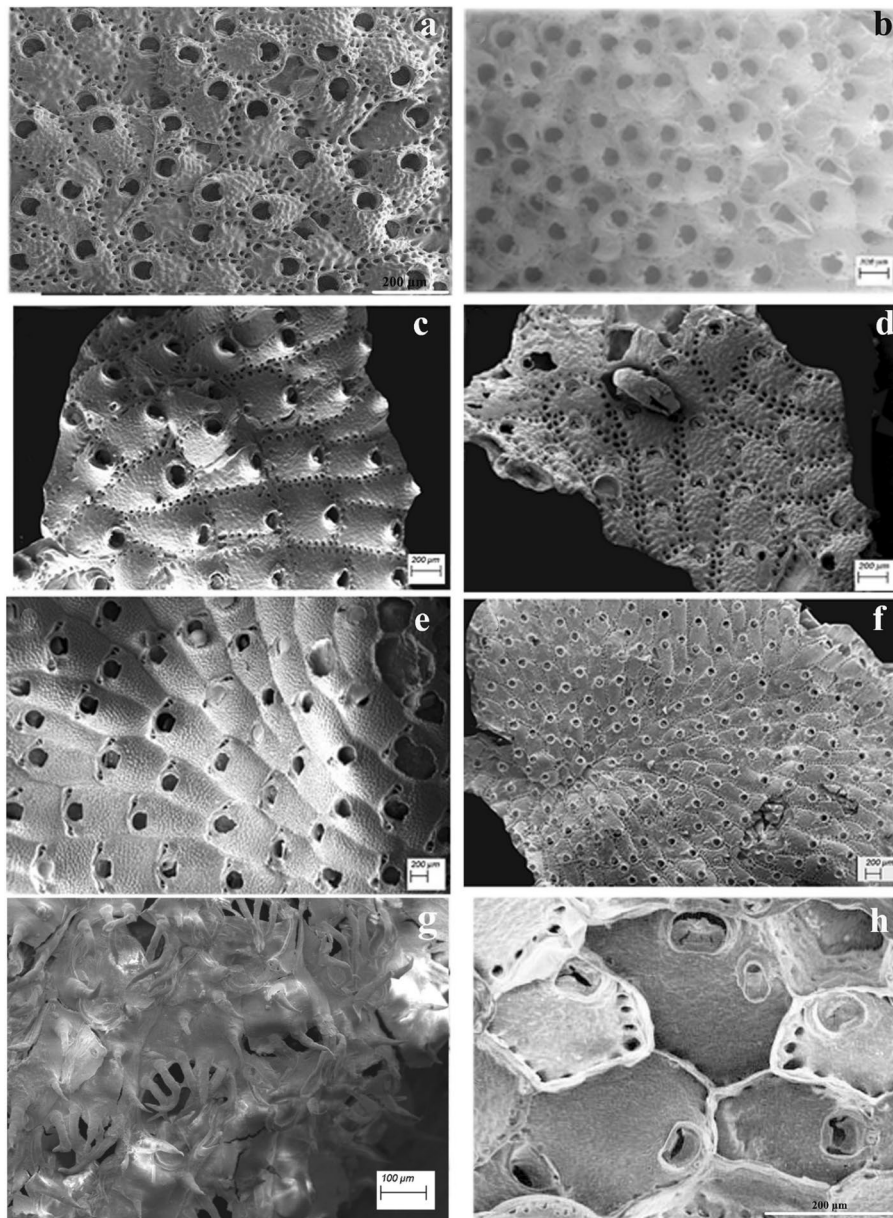


Fig. 4 Some of the species observed in the present study. **a:** *Parasmittina projecta* [36]; **b:** *Celleporaria aperta* [16]; **c:** *Parasmittina tubula* [22]; **d:** *Parasmittina egyptiaca* [59]; **e:** *Hippopodina feegeensis* [8]; **f:** *Pleurocodonellina signata* [58]; **g:** *Electra pilosa* [26]; **h:** *Parasmittina delicatula* [8]

3.1.1.2 Vedalai (site 2) The front side of the Ceramic plate which was smooth in texture, was completely coated by sediment deposits and no bryozoan colonies were noticed while the reverse side which had comparatively lesser sediment deposition, had one colony of *P. signata* and 16 colonies of *P. projecta*. Innumerable large colonies were noticed on the upper half of the plate (Fig. 5 VCF, VCR). The smooth textured front side of the Terracotta plate was completely coated by sediment deposits but no bryozoan colonies were noticed, while the reverse side of the plate which had a rough texture, was found to be encrusted by a single colony of *P. signata* and eight colonies of *P. projecta* but fewer sediment deposits were seen (Fig. 5 VTE, VTR). Both the surfaces

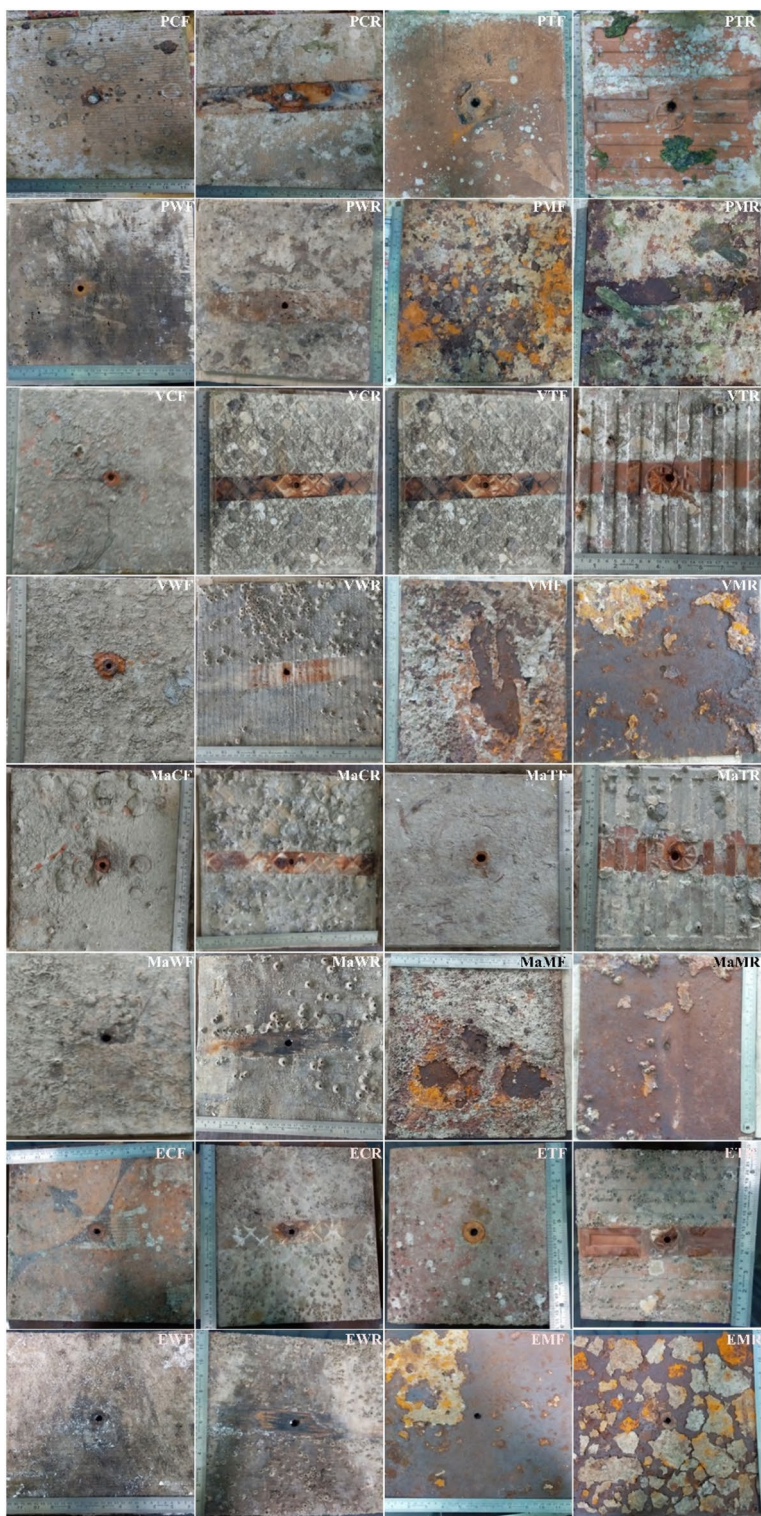


Fig. 5 Plates from short term study-1. P: Pamban, V: Vedalai, Ma: Manouli, E: Ervadi; C: ceramic, T: terracotta, W: wood, M: metal; F: frontal, R: reverse



Fig. 6 Plates from short term study-2. P: Pamban, V: Vedalai, Ma: Manouli, E: Ervadi; C: ceramic, T: terracotta, W: wood, M: metal; F: frontal, R: reverse

of wood were completely coated with sediment deposits and barnacles were found to be growing on the reverse side with higher concentration on the upper half of the wood. However, no bryozoan colonies were noticed (Fig. 5 VWE, VWR). The metal sheet which, was not painted, was highly rusted and had started to crumble. Although both the surfaces were completely coated with algae, salt and sediment deposits, no bryozoan colonies were noticed (Figs. 5 VME, VMR; 8i).

3.1.1.3 Manouli Island (site 3) The smooth textured front side of Ceramic plate was completely coated with sediment deposits and bivalves, but no bryozoan colonies were noticed. Barnacles were found to be growing on the rough reverse side which was completely coated with encrusting *P. projecta*, with higher concentration on the upper half of the plate along with six colonies of *C. aperta* (Fig. 5 MaCF, MaCR). The smooth textured front side of the terracotta plate was completely coated with salt deposits but no bryozoan colonies were noticed. The rough reverse side was found to have barnacles and bivalves growing on it along with 38 colonies of *P. projecta* and three colonies of *C. aperta*. A small percentage of the bryozoan colonies were also noticed on the barnacle shells. Higher concentration of fouling community was noticed on the upper half of the plate (Fig. 5 MaTF, MaTR). The front side of wood was completely coated with sediment deposits, on which a few barnacles and bivalves were growing. More barnacles were found to be growing on the reverse side with higher concentration on the upper half of the wood. No bryozoan colonies were noticed (Fig. 5 MaWF, MaWR). The metal sheet which was not painted was partly rusted on the front side and completely coated with sediment deposits while the reverse side was highly rusted and had started to crumble and fall apart. No bryozoan colonies were recorded (Figs. 5 MaMF, MaMR; 8i).

3.1.1.4 Ervadi (site 4) The front side of the ceramic plate which was smooth in texture, was partly barren with the exception of some algae, no bryozoan colonies were noticed. Barnacles were found to be growing on the rough reverse side with higher concentration on the upper half of the plate along with a single colony of *P. projecta* and three colonies of *C. aperta* (Fig. 5 ECF, ECR). The front side of the terracotta plate which was smooth in texture, was completely coated with salt and sediment deposits on which barnacles were growing but no bryozoan colonies were noticed. More barnacles were found to be growing on the reverse side with higher concentration on the upper half of the plate along with three colonies of *P. projecta* and three colonies of *C. aperta* (Fig. 5 ETF, ETR). Both the surfaces of wood were completely coated with algae, salt and sediment deposits with barnacles growing on them, but no bryozoan colonies were noticed (Fig. 5 EWF, EWR). The metal sheet which was not painted was highly rusted and had started crumbling apart but three colonies of *P. projecta* were noticed (Figs. 5 EME, EMR; 8i).

3.1.2 Short-term study 2 (January 2021–June 2021)

Careful examination of all the available plates submerged at four different sites in the study area after a period of 6 months revealed the presence of only four encrusting

bryozoan species. *P. projecta* and *C. aperta* were the most abundant among them but *P. projecta* was present on all plates irrespective of the substratum. *H. feegeensis* and *Parasmittina delicatula* [8] were observed very rarely, only as a small colony or a patch on the submerged panels (Fig. 4).

3.1.2.1 Pamban Island (site 1) The ceramic plate broke into pieces while detaching from the panel and only five pieces were able to be retrieved. One colony of *P. delicatula* was recorded and small barnacles and polychaete worms were also found along with it on the smooth reverse side of the plate. The rough surface was deposited with sand (Fig. 6 PCF, PCR). Polychaete worms and small barnacles covered the rough reverse side of the terracotta plate and a single colony of *P. delicatula* was found on the front smooth side (Fig. 6 PTF, PTR). The wooden block was completely degraded so was not analysed for the presence of bryozoans. The metal sheet was lost; therefore, it was not analysed for the presence of bryozoan (Fig. 8ii).

3.1.2.2 Vedalai (site 2) The Ceramic plate was broken into pieces while detaching from the panel and only three pieces were able to be retrieved. The smooth front side was devoid of any bryozoan colonies while the rough reverse side had 17 colonies, of which ten colonies were of *C. aperta* and six colonies of *P. projecta* and a single colony of *H. feegeensis* (Fig. 6 VCF, PCR). The terracotta plate broke into pieces while detaching from the panel but all the pieces of the plate were retrieved. The front smooth side was devoid of any bryozoan colonies but the rough reverse side had 22 colonies out of which 20 colonies were of *C. aperta* and two colonies of *P. projecta* (Fig. 6 VTF, VTR). Other than sand particles, both the surfaces of the wooden plate were devoid of any type of encrusting fauna (Fig. 6 VWF, VWR). The metal plate though rusted, was intact. A single colony of *C. aperta* observed on the reverse side while the front side was devoid of any bryozoan colony (Figs. 6 VME, VMR; 8ii).

3.1.2.3 Manouli Island (site 3) The rough reverse side of the ceramic plate had 12 colonies of *P. projecta* and 20 colonies of *C. aperta*, while the smooth front side was partly coated with sediments (Fig. 6 MaCF, MaCR). The terracotta plate was completely coated with sediments on both the front and reverse sides. The front side which was smooth in texture was devoid of any bryozoans while the rough reverse side had 15 colonies of bryozoa, of which 13 were colonies of *P. projecta* and two, colonies of *C. aperta* (Fig. 6 MaTF, MaTR). The wooden block was completely degraded. The reverse side of the metal plate contained 12 colonies of *P. projecta* mostly towards the upper half. On the other hand, the front side was completely coated with sediments and no bryozoan colony was visible (Figs. 6 MaMF, MaMR; 8ii).

3.1.2.4 Ervadi (site 4) Both sides of the ceramic plate were covered with thick layers of sediments. The rough reverse side was covered with 15 colonies of *P. projecta* and 20 colonies of *P. delicatula* most of these colonies were found on the upper half of the plate. The smooth front side was completely devoid of any bryozoan colony but had some large bivalves and a few barnacles attached on to it (Fig. 6 ECF, ECR). Both sides of the Terracotta plate were coated with thin layers of sediments. The rough reverse side had three colonies of *P. projecta* and three colonies of *C. aperta*, while the smooth

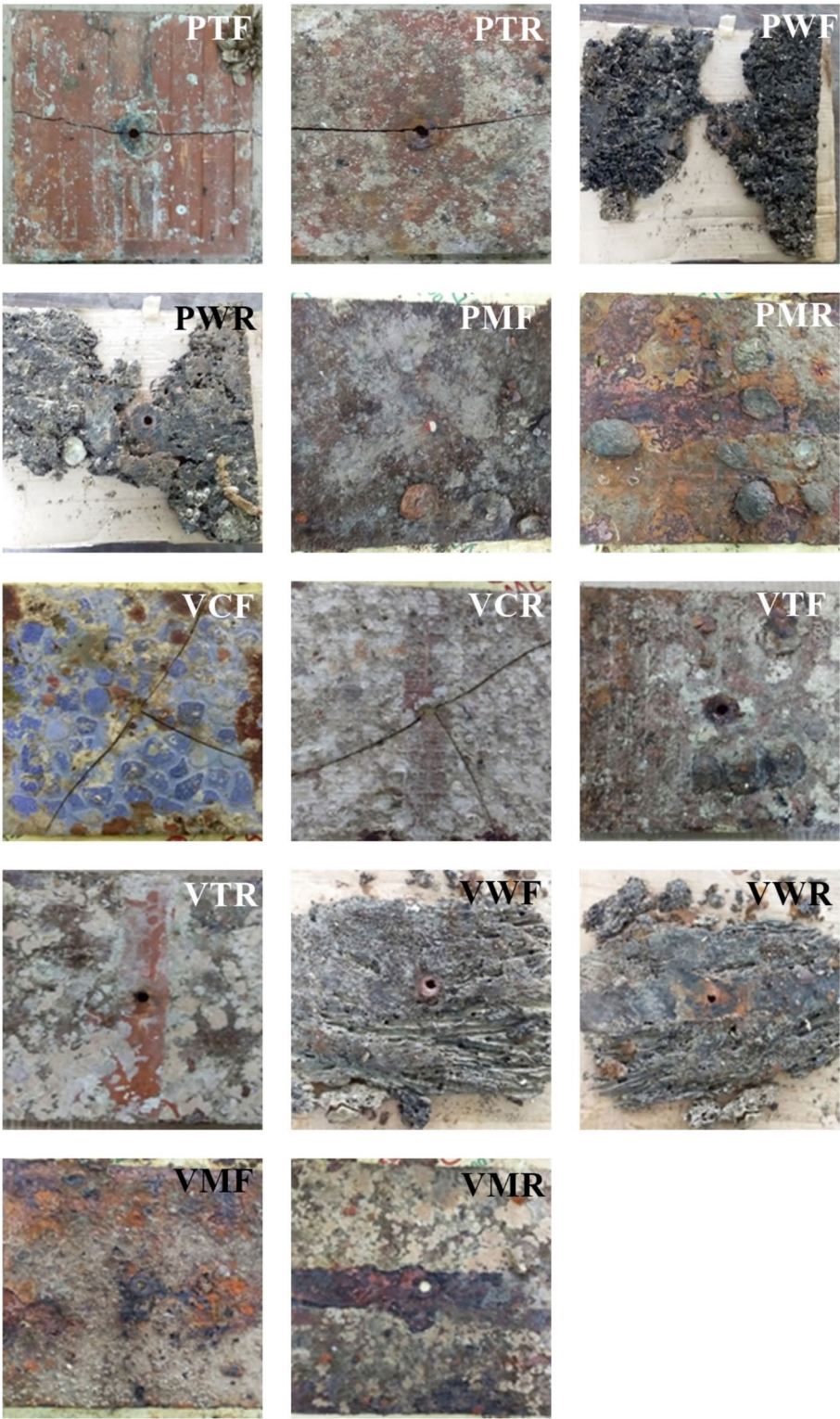


Fig. 7 Plates from short term study-2. P: Pamban, V: Vedalai, Ma: Manouli, E: Ervadi; C: ceramic, T: terracotta, W: wood, M: metal; F: frontal, R: reverse

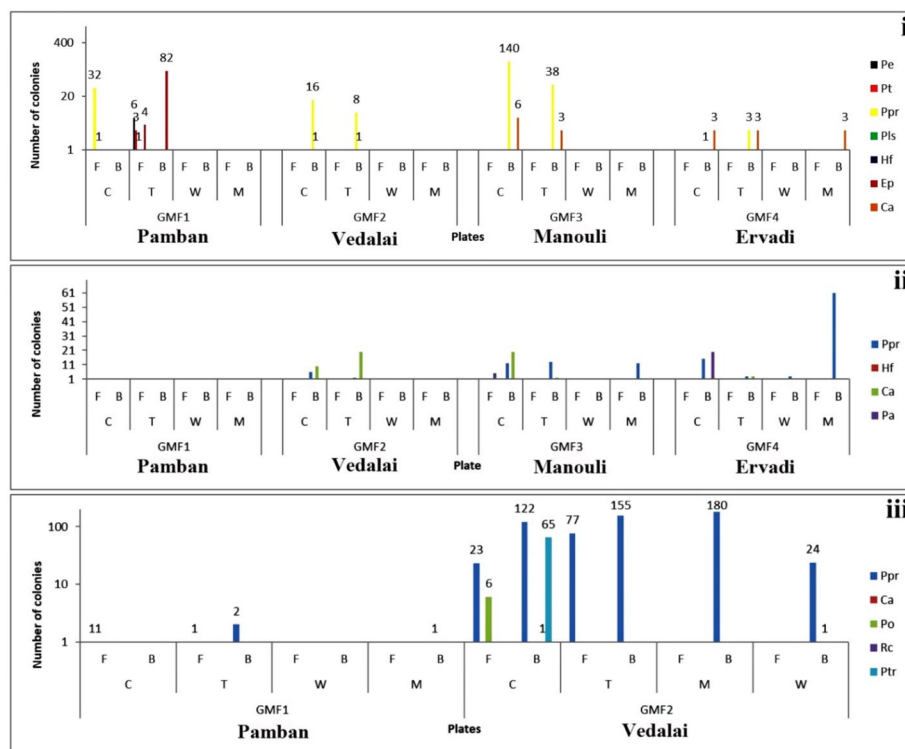


Fig. 8 Graphical representation of bryozoans settled during the studies. **i**: short term study-1, **ii**: short term study-2, **iii**: long term study; Substrate material: C—ceramic, T—terracotta, W—wood and M—metal; Side facing the swell: F—front and B—Back; Species: Pe—*Parasmittina egyptiaca* [59], Pt—*Parasmittina tubula* [22], Ppr—*Parasmittina projecta* [36], Hf—*Hippopodina feegeensis* [8], Ca—*Celleporaria aperta* [16], Pa—*Parasmittina delicatula* [8], Pls—*Pleurocodonellina signata* [58], Ep—*Electra pilosa* [26], Po—*Paralicornia oblecta* [15], Rc—*Rhynchozoon compactum* [52], Ptr—*Parasmittina triangularis* [29]

front side was devoid of any encrusting fauna (Fig. 6 ETF, ETR). The front side of the wooden plate was completely coated with thick layers of sediments while the reverse side was partially clean except for some sand and three colonies of *P. projecta* (Fig. 6 EWF, EWR). The front side of the metal plate, despite being rusted was not crumbled and was coated with sediments, but was devoid of any type of encrusting fauna. The reverse side was less rusted and 61 colonies of *P. projecta* were identified (Figs. 6 EMF, EMR; 8ii).

3.2 Long-term study of the bryozoans

Data of the bryozoans observed on panels submerged at two different sites in the study area during a period of 1 year revealed the presence of five species of encrusting bryozoans out of which *P. projecta* and *C. aperta* were observed abundantly on all substrata while, *Parasmittina triangularis* [29] and *Rhynchozoon compactum* [52] were observed only very rarely as a small colony or a patch on the submerged panels. Among the recorded bryozoans, *Paralicornia oblecta* [15] had been previously reported from Kovalam, on the South West coast of India by Menon and Nair [33], Indian Ocean [53] and Sri-Lanka [52] and from the present study site.

3.2.1 Pamban Island (site 1)

The Ceramic plate was lost; therefore, it was not analysed for the presence of bryozoans. The front side of the terracotta plate which was rough in texture was devoid of any encrusting fauna apart from some sea weeds growing on it. However, the reverse side which was smooth in texture was coated with salt and sediment deposits on which one colony of *C. aperta* and two colonies of *P. projecta* were encrusted (Fig. 7 PTE, PTR). The wood was highly degraded and infested by serpulid worms and bivalves. Some young crabs were also noticed, but no bryozoans were found (Fig. 7 PWE, PWR). The unpainted metal sheet was highly rusted but it did not crumble as it was completely covered on both sides by sea weeds, serpulid worms, barnacles and large bivalves. The reverse side was encrusted by a large colony of *C. aperta* (Figs. 7 PME, PMR; 8iii).

3.2.2 Vedalai (site 2)

The front side of the ceramic plate was encrusted with six colonies of *P. oblecta* and three colonies of *P. projecta*. The colonies were only at the corners and edges of the plate where the texture was roughened due to wear and tear. The back rough side was completely covered by encrusting bryozoans comprising of 122 colonies of *P. projecta* 55 colonies of *R. compactum* and 65 colonies of *P. triangularis* (Fig. 7 VCE, VCR). The front side of the terracotta plate which was rough in texture was infested with algae, barnacles, serpulid worms and bivalves along with 77 colonies of *P. projecta*. The smooth reverse side was completely coated with 115 colonies of encrusting *P. projecta* (Fig. 7 VTE, VTR). The wood was highly degraded and infested by serpulid worms and bivalves, some young crustaceans were also noticed. No bryozoans were found on the front side while 24 colonies of *P. projecta* and one colony of *P. oblecta* were noticed in the ridges and crevices on the reverse side of the rotten wood (Fig. 8 VWF, VWR). Both the sides of the metal sheet were heavily encrusted with algae, barnacles, serpulid worms and bivalves but 180 colonies of *P. projecta* were recorded on the reverse side (Figs. 7 VME, VMR; 8iii).

3.2.3 Manouli Island and Ervadi Island (site 3 and 4)

The plates could not be retrieved since the panels were lost.

3.3 Bryozoan percentage cover

Bryozoan settlement varied markedly among substrate types, panel orientations, and study periods. Ceramic and terracotta substrates consistently supported higher percentage cover than wood and metal, which showed minimal colonization throughout the study (Table 1). Shapiro–Wilk tests indicated non-normal distribution of percentage cover data ($p=0.024$); therefore, non-parametric analyses were applied. Kruskal–Wallis tests revealed significant differences in bryozoan cover among substrate types ($p<0.05$), with post-hoc comparisons showing significantly greater settlement on ceramic and terracotta substrates compared to wood and metal. Differences between ceramic and terracotta were comparatively small ($p=0.21$). Panel orientation significantly influenced settlement patterns, with back-facing surfaces exhibiting higher bryozoan cover than front-facing surfaces (Mann–Whitney U test, $p=0.013$). Seasonal comparisons showed increased settlement during the later study period,

Table 1 Bryozoan percentage cover on settlement plates

	Ceramic		Terracotta		Wood		Metal	
	Front	Back	Front	Back	Front	Back	Front	Back
Short term study 1 (June 2020 to January 2021)								
Pamban	32 ± 1.28	0 ± 0.32	27 ± 3.23	64 ± 1.23	0 ± 0.32	0 ± 0.43	0 ± 0.43	0 ± 0.12
Vedalai	0 ± 0.67	23 ± 3.29	0 ± 1.29	18 ± 0.23	0 ± 0.67	0 ± 0.47	0 ± 0.57	0 ± 1.10
Manouli	0 ± 1.13	42 ± 2.34	0 ± 2.19	42 ± 2.18	0 ± 1.14	0 ± 1.19	0 ± 1.14	0 ± 0.34
Ervadi	0 ± 0.29	21 ± 2.12	0 ± 1.18	14 ± 0.89	0 ± 1.18	0 ± 0.94	0 ± 1.63	2 ± 0.66
Short term study 2 (January 2021 to June 2021)								
Pamban	8 ± 0.23	0 ± 0.48	0 ± 0.71	4 ± 0.63	0 ± 0.89	0 ± 0.15	0 ± 0.21	0 ± 0.13
Vedalai	0 ± 0.89	24 ± 0.69	0 ± 0.84	32 ± 1.03	0 ± 0.65	0 ± 0.83	1 ± 0.43	4 ± 0.64
Manouli	12 ± 0.34	37 ± 2.23	9 ± 1.93	23 ± 2.16	0 ± 1.18	0 ± 0.74	0 ± 1.27	13 ± 1.23
Ervadi	0 ± 1.09	41 ± 3.21	0 ± 0.36	18 ± 1.23	0 ± 0.96	7 ± 0.27	0 ± 0.48	43 ± 1.86
Long term study 2 (1 year)								
Pamban	54 ± 1.93	0 ± 1.23	28 ± 1.68	39 ± 1.37	0 ± 0.91	0 ± 0.62	0 ± 0.12	12 ± 0.29
Vedalai	67 ± 1.21	78 ± 1.67	42 ± 2.17	65 ± 1.12	0 ± 1.21	38 ± 1.24	0 ± 0.23	32 ± 1.27

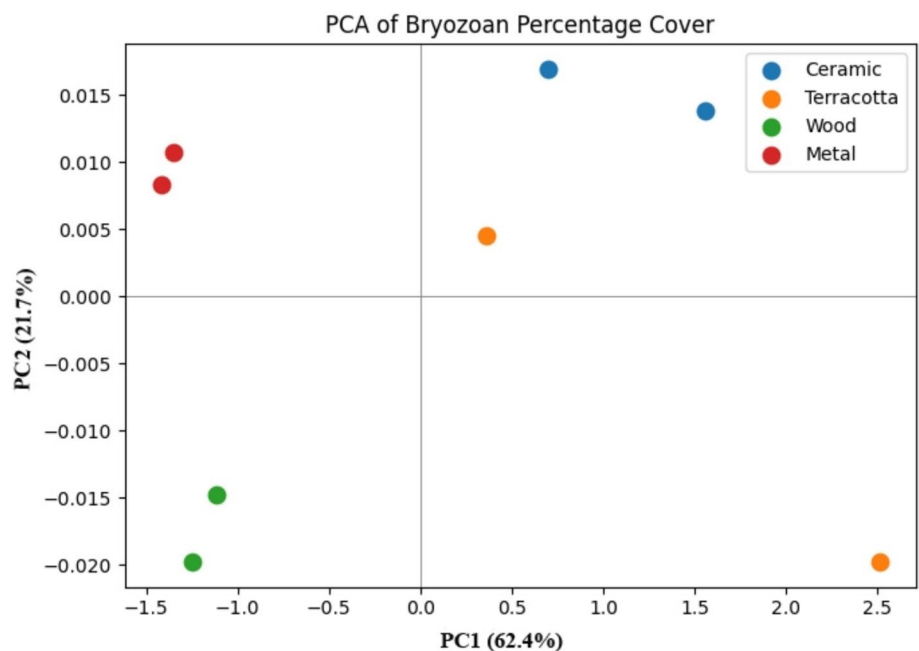


Fig. 9 Principal Component Analysis (PCA) of total bryozoan percentage cover across different substrate types and study periods

indicating temporal variation in recruitment. Principal Component Analysis (PCA) revealed clear separation of samples based on substrate type. The first principal component distinguished high-colonization substrates (ceramic and terracotta) from low-colonization substrates (wood and metal), while PC2 likely reflects combined variation associated with both seasonal and site-specific differences in bryozoan settlement patterns. Overall, PCA supported the univariate results and highlighted substrate-driven structuring of bryozoan settlement patterns (Fig. 9).

4 Discussion

4.1 Environmental context influencing bryozoan settlement

As tropical regions are strongly influenced by meteorological factors [5], the spatial settlement of these invertebrate species on the panels is unlikely to be random. Instead, their distribution and settlement appear to be shaped by their responses to a range of environmental cues and weather patterns. The present study attempted to analyse the settlement preferences of marine bryozoans related to season and substrate between June 2020 to June 2021 at four different locations in Gulf of Mannar.

Among all the physical and chemical parameters of water, temperature and salinity play a critical role in the wellness and vitality of reef ecosystems. Gopinathan and Rodrigo [14] reported that tropical aquatic organisms including shell and fin fishes reach their optimum growth at water temperatures ranging between 25 °C and 30 °C. Geetha and Kumar [12] also noted that coral reefs typically occur in areas with salinity levels of 32 to 40 ppm. At the present study site, salinity during the 2020 coral bleaching event was recorded between 35.0 and 35.5 ppm [45]. Drastic thermal shifts and wide salinity fluctuations can adversely affect reef building organisms, and can have a cascading effect on the entire reef system.

In addition, calcium which is essential for reef building organisms was also found in high concentrations during monsoon and at a lower concentration during the pre-monsoon, post-monsoon and summer season [12, 45]. Since CO₂ and calcium in seawater are vital for forming the calcium carbonate skeletons built by these natural reef engineers, they can only flourish when environmental conditions support this process, as was in the case during Study Period 1. This can explain the higher number of bryozoan colonies in study 1.

Water temperature remained within the optimum range (26 °C–28 °C) for most of the year, (June–March), with only a slight increase during summer reaching 29.9 °C–30.1 °C in the study area [12]. However, in 2020, a different pattern was observed, with temperatures ranging from 32.0 °C to 35.1° over the same period [45]. Additionally, a decrease in salinity was observed during the monsoon season (October to December), likely due to the substantial influx of freshwater entering the coastal waters at that time.

The higher salinity levels observed during the pre-monsoon, post-monsoon and summer season are likely the results of increased surface water evaporation combined with upwelling processes [4, 23]. The rise in water temperature and salinity noted during (January to May) study 2 period [45], though unfavourable for reef builders, are favourable conditions for corrosion of metal plates. Besides this, as the sea was less turbulent, anthropogenic activities were high, which could have also led to the damage and loss of plates. These could be the reasons for the significant drop in the number of bryozoan colonies recorded during the study 2 period.

4.2 Substrate type as a determinant of bryozoan settlement

The substrate type and availability of space on the substratum are amongst the most important environmental factors controlling the distribution of bryozoans. The present study explores the substrate preference of bryozoans to ceramic, terracotta, wood and metal commonly found on the hulls of boats and ships or on docs and harbours.

Bryozoans showed a preference to terracotta and ceramic when compared to wood and metal. Woollacott [60], Rittschof and Costlow [43] demonstrated that 'wettability', an indication of surface hydrophilicity, is higher in terracotta and ceramic than in wood and metal. The absence of bryozoan colonisation on metal panels may be attributed to corrosion-induced surface instability and the release of iron oxides, which are known to negatively influence larval settlement and early colony development of sessile invertebrates [40]. The substrate texture was yet another factor of prime importance that influenced larval settlement leading to the formation of mature colonies [44]. Although both terracotta and ceramic plates had rough and smooth surfaces, a greater number of colonies were noted on the rough surface of the ceramic plate than on the smooth surface, while no such texture preference was observed on the terracotta plate.

4.3 Effects of panel orientation, shading, and hydrodynamics

The reverse sides of the panels were comparatively more shaded and sheltered, conditions that are known to favour bryozoan settlement. Many encrusting bryozoans are sciaphilic, with larvae preferentially settling in low-light or shaded microhabitats where algal competition and physical disturbance are reduced [41, 44, 60]. This ecological preference likely contributed to the consistently higher abundance of bryozoan colonies observed on the reverse surfaces of the panels.

In addition to shading, the 45° inclination of the panels influenced local hydrodynamic conditions, resulting in a non-uniform distribution of colonies along the panel surfaces. The upper portions of the panels experienced greater water movement, which can enhance the supply of larvae and suspended food particles without necessarily increasing sediment deposition. Such moderate flow conditions are favourable for suspension-feeding encrusting organisms, whereas excessive sediment accumulation may impair feeding by clogging the lophophores. Similar effects of panel orientation on settlement patterns have been reported previously [24].

Furthermore, panel orientation mediated exposure to wave action and predation. The front surfaces were subjected to stronger hydrodynamic stress and higher visibility to potential predators, while the reverse surfaces experienced comparatively reduced wave impact and predation pressure. This differential exposure likely promoted greater settlement success and survival of bryozoan colonies on the reverse sides, consistent with commonly observed patterns in sessile marine invertebrates.

4.4 Seasonal and temporal variation in bryozoan settlement

A comparative analysis done on the short-term study 1 and study 2 revealed the presence of seven different species on the plates of short-term study 1 while only four species were found on the study 2 plates. *H. feegeensis*, *C. aperta* and *P. projecta* were common in both study 1 and study 2, while *P. egyptiaca*, *P. tubula*, *P. signata* and *E. pilosa* were only found on study 1 plates and *P. delicatula* was found on study 2 plates alone. However, bryozoan presence was only found on ceramic and terracotta plates in study 1 at the same time, all the four plates seemed to have bryozoans in either one or other location in study 2. Comparative Graphical representation of bryozoans settled on tiles during short term study-1 and study-2 is represented in (Fig. 10i).

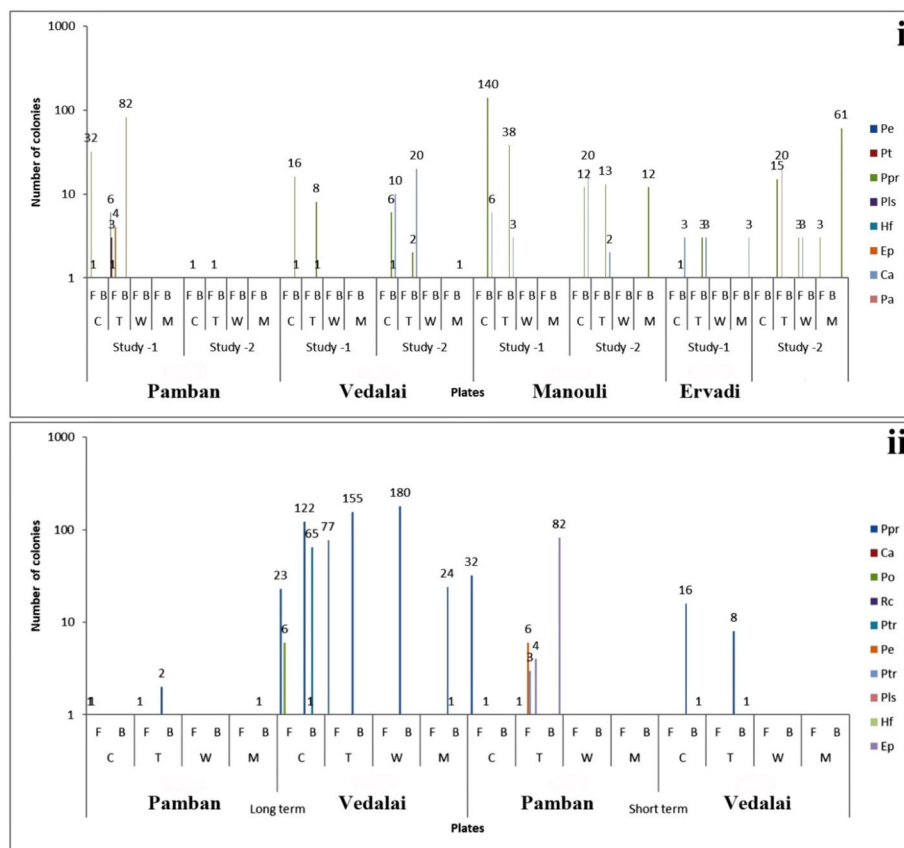


Fig. 10 Graphical representation of comparison of bryozoans settled (total number of colonies across all studies and areas) during **i**: short term study 1 vs short term study 2, **ii**: long term study vs short term study; Substrate material: C—ceramic, T—terracotta, W—wood and M—metal; Side facing the swell: F—front and B—Back; Species: Pe—*Parasmittina egyptiaca* [59], Pt—*Parasmittina tubula* [22], Ppr—*Parasmittina projecta* [36], Hf—*Hippopodina feegeensis* [8], Ca—*Celleporaria aperta* [16], Pa—*Parasmittina delicatula* [8], Pls—*Pleurocodonellina signata* [58], Ep—*Electra pilosa* [26], Po—*Paralicornia obtecta* [15], Rc—*Rhynchozoon compactum* [52], Ptr—*Parasmittina triangularis* [29]

4.5 Species-specific patterns and successional dynamics

The present study focused on the texture and substrate preference of bryozoans to ceramic, terracotta, wood and metal at two different sites in the study area during a period of 1 year. The results revealed ten species of encrusting bryozoans in the Gulf of Mannar, acting as fouling organisms. *P. projecta* was found to be highly dominating on most of the panels irrespective of the substratum and time period, as they were identified on all substrata, both on short term panels and on long term panels alike. *P. triangularis*, *P. tubula*, *P. egyptiaca*, *H. feegeensis*, *R. compactum* and *P. signata* were observed only very rarely as a small colony or a patch on the submerged panels. *P. tubula*, *P. egyptiaca* and *H. feegeensis* were found only during short term study, while *P. triangularis* and *R. compactum* were found only during long-term study. *E. pilosa* was only found on the rough surface of the terracotta plate at site 1 in the short-term study, while *P. obtecta* only on the rough surface of the ceramic plate at site 2 in the long-term study (Fig. 10ii). It was also observed that a competent species of bryozoans grows over the less competent species resulting in the displacement of the latter one (Fig. 11). The panels that

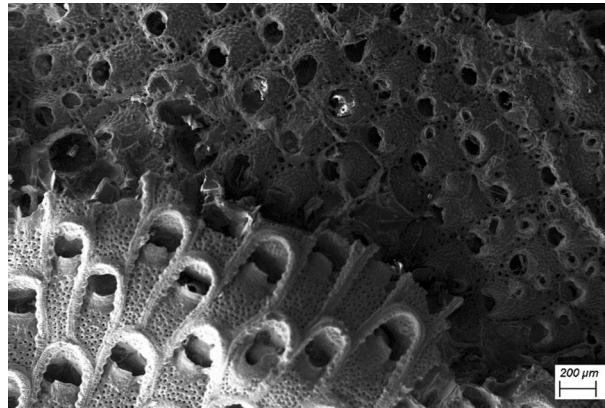


Fig. 11 SEM image of *Celleporaria aperta* Hincks [16] growing over a less competent species

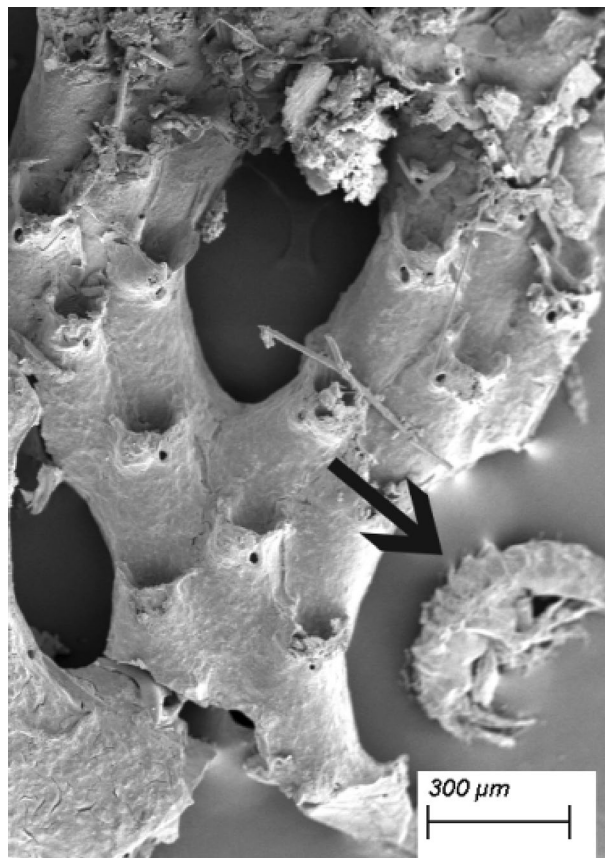


Fig. 12 SEM image of a crustacean that took shelter among the bryozoan (*Hippopodina feegeensis* Busk [8])

had been submerged for about a year, harboured animals belonging to the category of hydroids, serpulid worms and barnacles along with thickets of sea weeds which transformed into a microhabitat on the artificial substratum providing shelter for juveniles of crabs and other crustaceans (Fig. 12). This was not observed on the panels submerged for 6 months.

The data collected here, clearly demonstrate the role bryozoans play within the marine biofouling community. Eleven species of encrusting bryozoans in the Gulf of Mannar act as fouling organisms, among which *P. obtecta* is newly reported from the area. However, it had previously been reported from Kovalam, in the Southwest coast of India [33] as well as from the Indian Ocean [53] and Sri Lanka [52]. All bryozoan species observed on the panels had previously been reported as epifauna on algae, sponges, serpulid worms, bivalves, corals or on rocks, but this is the first time they have been recorded in this area on artificial substrates.

P. projecta and *C. aperta*, were identified on all substratum, both on short term panels (study 1 and study 2) and on long term panels. This may be due to their greater tolerance to a wider range of physical and environmental factors which allows them to behave like pioneer organisms in facilitative successions [35, 60]. *P. triangularis*, *P. tubula*, *P. egyptiaca*, *H. feegeensis*, *R. compactum*, *P. delicatula* and *P. signata* were infrequently found on the submerged panels from both long-term and short-term studies, appearing only as small colonies or patches. Although Bryozoa compete with other encrusting organisms on hard substrates, they can be overgrown as competition increases during succession progresses. This may explain why these species are rarely seen and occur only as small colonies, as also noted by McCloskey [30], Enochs [10], Barnes and Neutel [6] and Sanjay et al. [46]. *P. tubula*, *P. egyptiaca* and *E. Pilosa* were found only in the short-term study 1 and may be due to the favourable conditions experienced during short term study 1 period. In contrast, *P. delicatula* was found only in the short-term study 2, and may be due to its halotolerant nature, which aligns with the findings of Menon and Nair [33, 34] demonstrating the influence of salinity on fouling marine bryozoans.

An interesting observation noted is that *P. triangularis* and *R. compactum* were found only on the long-term study panels and not in either short term study 1 or study 2. The restriction of these two species to the long-term immersion panels suggests that these taxa can represent later successional species within the biofouling community. In addition to successional status, larval settlement of these species may be influenced by the development of a specific or mature microbial biofilm, which forms progressively over time. It is well established that biofilm composition, thickness, and microbial diversity change with immersion duration and can provide chemical and tactile cues essential for the settlement of certain bryozoan larvae [28, 61]. On the other hand, *H. feegeensis* was found in both short-term study 1 and study 2 but not in the long-term study. This could be due to them being overgrown during competitive interactions over the course of the long-term study, a pattern also observed by Enochs [10].

The formation of a primary biofilm and subsequent attachment of macrofoulers, including bryozoans, leading to a climax community was clearly observed in the present study. This aligns with Scheer [47] who found that fast growing serpulids, hydroids, bryozoans and ascidians shape the climax community. Similar observations were made by Menon et al. [32], Braiko [7], and Khalaman [21] in several established ports in south India. Panels submerged for 1 year were colonized by hydroids, serpulid worms and barnacles along with dense growths of seaweed, which transformed the plates into microhabitats providing shelter for juvenile crabs and other crustaceans. Macrofoulers and seaweeds were more abundant on the panels submerged for 1 year compared to the panels submerged for 6 months. Most studies on fouling

community succession have been conducted near the coast or shallow waters, where succession is often disrupted by unstable environment such as variable salinity, temperature, light levels and depth, physical disturbance, pollution [41]. In contrast, the present study was carried out in shallow coastal waters or nearshore waters at a depth of 5 m depth, where physical disturbances are relatively lower, allowing clearer observation of macrofouling succession and climax community establishment.

4.6 Integration of univariate and multivariate results

From the bryozoan percentage cover data, it has been observed that bryozoan settlement was strongly structured by substrate type, panel orientation, and study period, with substrate emerging as the primary determinant of percentage cover. Ceramic and terracotta consistently supported significantly higher bryozoan cover than wood and metal, a pattern widely reported in settlement-panel studies and attributed to material-specific surface properties and associated biofilm development that influence larval attachment and retention [38, 49]. Bryozoan larvae are known to exhibit substratum-selective settlement behaviour, responding to physical and microbial cues that can amplify material-driven differences into pronounced contrasts in realized cover [11, 13]. Conversely, persistently low cover on wood and metal may reflect reduced larval attraction and/or elevated early post-settlement mortality, consistent with recruitment theory emphasizing strong post-settlement filtering [17].

Panel orientation further influenced settlement, with higher cover on back-facing surfaces, likely reflecting reduced sedimentation, hydrodynamic stress, or post-settlement loss on sheltered faces, as reported in other fouling studies [49, 57]. Decreased settlement during the later study period (study period 2) indicates temporal variability in recruitment, consistent with documented seasonal effects driven by changes in temperature, food availability, and reproductive timing in bryozoans [51]. PCA results integrated these patterns, highlighting substrate-driven structuring with a secondary temporal axis, reinforcing the dominant role of material type in early bryozoan colonization on artificial substrates.

4.7 Implications for biofouling research in the Gulf of Mannar

The findings of the present study contribute substantially to the limited body of experimental biofouling research in the Gulf of Mannar. While previous Indian studies have largely focused on harbour environments [31, 33, 34], our shallow coastal water analysis provides the first evidence of substrate-specific bryozoan succession in this biosphere reserve. The dominance of *P. projecta* and *C. aperta* is consistent with global fouling studies [10, 44], reinforcing their role as pioneer, fast-growing taxa capable of creating the foundational structure for macrofouler communities. The study further aligns with international literature emphasizing the ecological engineering role of bryozoans in reef stabilization and habitat formation [9]. The observations from the present study support existing knowledge on the role of bryozoans as reef-modelling and habitat-forming organisms, and provide region-specific evidence from the Gulf of Mannar.

4.8 Limitations

The present study, while providing novel insights into bryozoan biofouling and succession in the Gulf of Mannar Marine Biosphere Reserve, has certain limitations that should be acknowledged.

First, the experimental design did not include true replication. At each site, one short-term (6-month) and one long-term (1-year) immersion panel were deployed, representing distinct temporal exposures rather than replicated units. Consequently, the results are interpreted qualitatively and comparatively, focusing on substrate preference, settlement patterns, and successional trends rather than statistical inference. The absence of replication was primarily due to logistical constraints associated with underwater deployment, limited dive windows, and regulatory restrictions within a protected biosphere reserve. Also due to the absence of true replication, parametric statistical analyses were not performed, and results are interpreted descriptively and comparatively.

Second, some panels and plates were lost or degraded during the course of the experiment. These losses occurred mainly due to strong monsoonal currents, prolonged exposure to wave action, corrosion and crumbling of unpainted metal substrates, biofouling-induced structural stress, and occasional anthropogenic disturbances during calmer sea conditions. While additional anchoring or protective frames may reduce losses, such challenges are common in long-term open-water biofouling experiments and reflect realistic environmental conditions rather than experimental failure.

Third, environmental parameters were not measured in situ during the study period. Continuous measurements of variables such as temperature, salinity, turbidity, current velocity, dissolved oxygen, and chlorophyll-a concentration would have strengthened interpretations of seasonal settlement patterns. In particular, chlorophyll-a could have served as a proxy for primary productivity and plankton availability, which are known to influence larval survival and recruitment success of suspension-feeding organisms such as bryozoans [27]. Future studies integrating environmental sensors with replicated panel deployments would allow a more mechanistic understanding of biofouling dynamics.

Despite these limitations, the study offers valuable baseline data from an understudied region and highlights practical challenges inherent to long-term marine colonisation experiments, providing useful guidance for the design of future biofouling studies in open-water reef environments.

5 Conclusion

Bryozoans form a solid skeletal crust that covers surfaces, providing a substrate for the attachment of macrofoulers. As such, they act as ideal pioneer species, initiating secondary succession through facilitation, followed by competitive succession, ultimately leading in a climax community composed of hydroids, serpulids, barnacles, ascidians and bryozoans. The present study highlights the positive role of biofouling bryozoans, demonstrating their potential to facilitate the settlement and succession of macrofoulers, thereby contributing to the formation of community structure and biodiversity [37, 48]. Consistent with global observations, bryozoans function as important habitat-forming organisms by creating structural complexity and facilitating the establishment of

diverse macrofouling assemblages on submerged surfaces. Additionally, this study provides region-specific baseline data on bryozoan fouling dynamics on artificial substrates, which may inform understanding of community development on man-made marine structures and guide future targeted research on reef-associated interactions. Such artificial structures offer opportunities for fisheries, underwater recreational activities and ecotourism without disturbing endangered natural reefs, and they can also contribute to disaster mitigation by resisting the destructive forces of waves during events such as hurricanes and tsunamis.

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Author contributions

MSS—Data curation, Formal analysis, Investigation, Writing—original draft; VC—Conceptualization, Methodology, Supervision, Validation, Writing—original draft; AS—Software, Writing—review and editing; YKJS—Formal analysis, Methodology, Investigation, Writing—review and editing.

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Conflict of interests

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Ethics approval

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Consent to participate

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